

Consolidated Findings: Simple Policies, Information, and the Room for RL in the Pharmaceutical Supply-Chain Simulator

One running document consolidating the routing study, the fixed-routing (understanding) study, and the value-decomposition study (Phases 1–2, DS-seat test, Part C). Individual reports remain archived as the audit trail.

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1 Current state of claims

Every load-bearing claim, its current value, and whether it superseded an earlier number. All values from audited runs on held-out reporting seeds unless marked.

Claim	Current value	Confidence	Supersedes
Disruption pile-up is a routing artifact, not physics	67–91% of cost removable by HC routing rules alone (thesis severity)	high	“pile-up unavoidable”
Info-sharing reroute = perfect-onset oracle	bit-identical trajectories; 1.06M→295k; lost 445→203	high (audited bit-equality)	—
Downstream detection is structurally blind	~6-period stock-cover masking (~0.9M); explicit detector worse than built-in trust	high	“detection captures ≈0” (narrowed: detect-then-buffer only)
True foresight value (slack)	~54k (~5%), with correctly LOCATED buffer	high	“foresight adds nothing”
Room above best deployable (slack)	≈0 <i>within the enumerated rule family</i> ; no candidate mechanism outside it	moderate-high (family bound)	absolute “room ≈0”
DS allocation channel, routing-fixed worlds	≤12% (static priority, fairness-costed); smart rules backfire +25–56%; priority flips HARMFUL under scarcity (+23%)	high	—
Demand-shaping from the DS seat (thesis world)	REAL: shed +21.8%, both HCs better; direction control −14.3%	high (audited)	“allocation ≤12% everywhere” (was rung-c-specific)
DS-seat compound vs base stock (thesis world, thesis metric)	49.1% (u0) / 50.7% (u20) of BS loss removed; fill 0.858; lost 1,575 vs 1,768	high	—
Thesis RL advantage (Table 3.9)	claimed 89% of BS loss; UN-REPLICATED; baseline config unknown	open	—
Taper (order-toward-failed-source)	+24.9% on thesis metric but dead-factory bookkeeping; +9% lost patients ALONE	high	—
State-elevated base stock	dead by supply physics (inv. cannot climb during outage: 4 units measured)	high	—
Scarcity frontier (sat50, d48)	compound (buffer-at-survivor + reroute + taper): −79%, fill 0.99 (1.00 at larger B); dominates buffer-only 2.4× at matched premium	high	fill “1.000” at B960 (rounding; 0.990)
Lever-flip map	no-action ($d \leq$ stock cover) → routing → compound → buffer-only	high (single-event regime)	sat30 cell: buffer belongs at DISRUPTED chain (2,417k, −30% vs healthy)
Demand-aware buffer sizing	B = (demand − surviving deliverable) × duration: lost 41 vs 505 at frozen optimum	high	frozen per-severity grids
Severity-aware redirect caps	REFUTED both forms (orders are the supplier’s production signal)	high	—
Buffers in slack	location-specific: at disrupted chain zero-to-negative; at surviving chain +4–7%	high	blanket slack-buffer negative; “insurance captures 42%” (broken-routing artifact)
Earlier value split (−468k buffer / −654k timing)	RETRACTED (measured on broken routing layer)	—	itself

2 The four findings that matter

F1 — The binding constraint is information, not intelligence. At thesis severity the surviving manufacturer has $3.3\times$ headroom and ramps with orders; a two-line reroute on the *shared upstream machine-state* signal exactly reproduces perfect onset knowledge. Without that sharing, NO downstream policy — learned or not — can act in time, because the dead distributor’s own stock masks the failure for ~ 6 periods. *Meaning: the thesis’s information-sharing narrative is right, but its value is captured by a trivial rule, not by a learner.*

F2 — The original hypothesis, half-confirmed, hinging on one missing file. From the distributor’s seat, in the thesis’s own world and metric, a 3-parameter compound (serve-the-captive $\times$ taper $\times$ standing buffer) removes 49–51% of base stock’s loss — against the RL’s claimed 89% from a table we cannot replicate (our own corrected-pipeline DRL *lost* to base stock). *Meaning: simple rules beat base stock decisively; whether the RL’s remaining margin is real is now a question about the thesis baseline configuration — the single most valuable item to obtain.*

F3 — Under scarcity, structure wins: location and sizing, not reaction. The frontier is owned by one composition; the buffer’s optimal *location* flips to the disrupted chain when the surviving chain is weak; and demand-aware *sizing* cuts lost patients from 505 to 41. *Meaning: the two deployable decisions that matter are where stock sits and how it is sized — both computable from observables, no learning required.*

F4 — Cleverness consistently loses to structure. Across three studies, every reactive “smart” rule backfired: delivery-rate detection, backlog-priority, serve-captive (in the routing-fixed world), rotating priority, priority-with-floor, and both severity-aware redirect caps. The trust feedback loop rewards consistency; the production loop is driven by orders. *Meaning: for the eventual RL comparison, the honest prior is that adaptivity must overcome a measured pattern of adaptive rules underperforming static structure.*

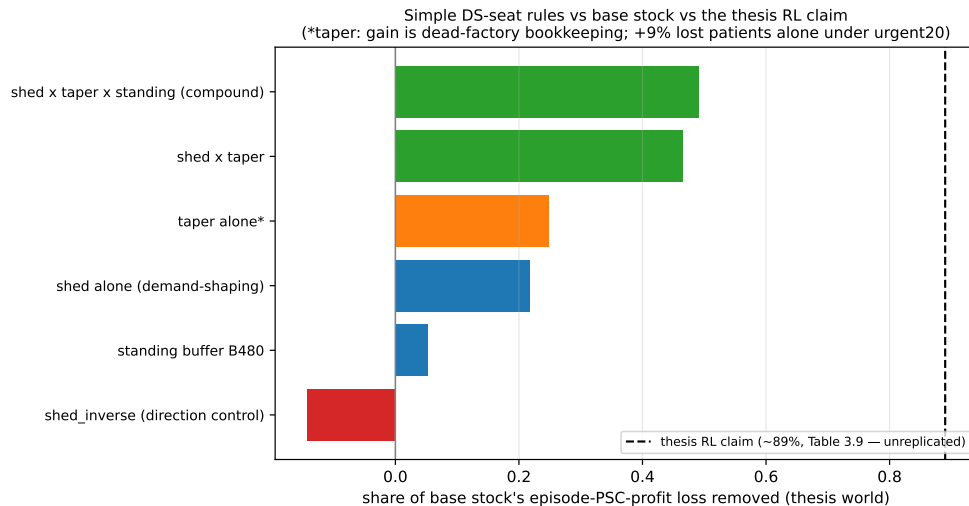


Figure 1: F2 in one picture: simple distributor rules vs the RL claim, thesis world, audited reporting seeds.

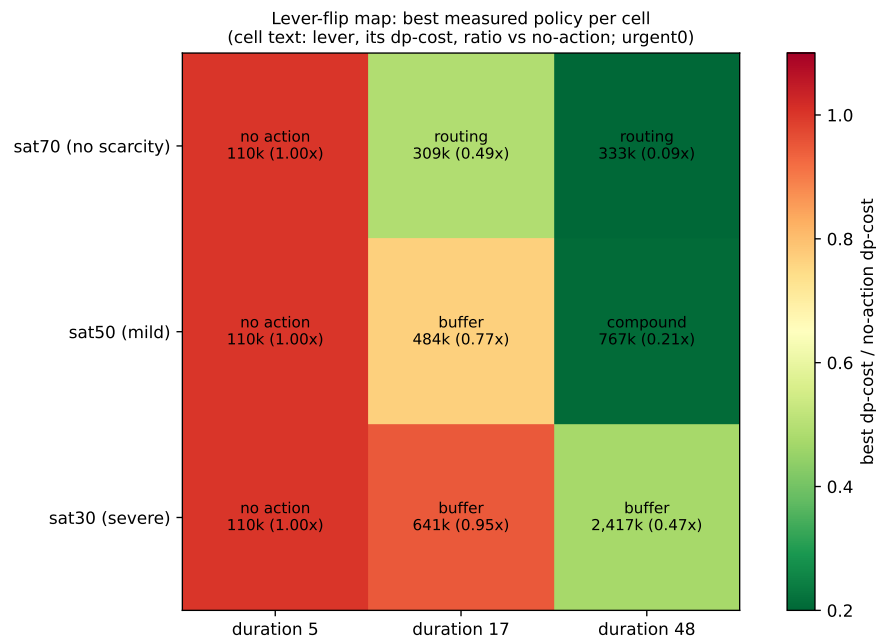


Figure 2: F3 in one picture: the lever-flip map (sat30 cell corrected: disrupted-located buffer). Pre-publication caveat: single-event episodes; recurrence could move buffer-cell boundaries (premium amortization) — flagged as the required robustness check.

3 Phase-by-phase detail

3.1 Background: the DRL arc (why there is no working RL comparator)

The reward audit found r1/r4 carrying a thesis-level algebraic defect, r5 a code defect, r2/r3/r6 saturated as configured; repairs un-stuck all six components. Training instability traced to the adaptive exploration rule (multiplicative runaway to its cap); a linear schedule yields a stably-training agent — which then *loses* to base stock on system profit (-4.06M vs -2.35M , driven by manufacturer queue cost). Table 3.9 is therefore treated as a claim, not a reproduced result, pending the author’s configuration.

3.2 Routing study (thesis severity)

HC2’s hard-wired equal-split plus stranded-order suppression lock $\sim 2/3$ of the disruption cost into the routing layer: trust-split -26% , sharpened trust + stale-order write-off -67% (fill 1.000), oracle onset reroute -91% ; lost patients $1,768 \rightarrow 203$. Found en route: the normal demand model hard-reseeds the global RNG (all historical multi-seed normal-demand results were one path) — fixed in our runners, gate-guarded.

3.3 Understanding study (routing-fixed world)

Allocation bounded at $\leq 12\%$ via max-over-family (static priority; fairness cost ± 0.023 fill); all dynamic rules backfire through the trust loop. The rung-c residual is 41% dead-factory queue book-keeping (dual accounting introduced for exactly this), the ~ 10 -period redirection transient, and a write-off glut later removed as a side effect of instant rerouting.

3.4 Value decomposition, Phase 1 (slack)

Info-sharing reroute $\equiv$ oracle (bit-identical, audited); explicit delivery-rate detection worse than the built-in trust mechanism (stock-cover masking $\sim 0.9\text{M}$); deployable compound 273k. Verification rebuilt the oracle as a true superset and corrected two claims: foresight is worth $\sim 54\text{k}$ with the buffer at the SURVIVING chain, and slack buffers are negative only at the disrupted location ($+4-7\%$ at the survivor).

3.5 Value decomposition, Phase 2 (saturated regimes)

With the surviving manufacturer also cut, routing still reduces cost ($-26\%/-50\%$) but cannot create product (lost patients $\sim$ flat across the entire ladder); the compound reaches -79% at fill 0.99–1.00; static priority flips harmful ($+23\%$); the severity $\times$ duration map has two boundaries (stock-cover duration; surviving-chain headroom), with sharp rerouting the WORST measured policy at severe scarcity.

3.6 DS-seat study (thesis world) and Part C

Full-grid ladder over the distributor’s complete decision space (everything else is dead by topology). Results in F2/F4 and the claims table. Part C: demand-aware sizing (the robust knob), the sat30

location correction, and the redirect-cap refutation with its mechanism (orders are the production signal).

4 Superseded claims (do not cite)

- The $-468\text{k} / -654\text{k}$ value split — measured on the broken routing layer; retracted.
- “Permanent insurance captures 42%” — broken-routing artifact; in the fixed-routing slack world, disrupted-located buffers are zero-to-negative (surviving-located: $+4\text{--}7\%$).
- “Foresight adds nothing beyond the shared signal” — artifact of a mis-located ceiling buffer; corrected to $\sim 54\text{k}$ ($\sim 5\%$).
- “Detection captures ≈ 0 ” — true only for detect-then-buffer; rerouting responses carry most of the onset value, and the upstream-signal trigger captures it fully.
- “Allocation $\leq 12\%$ is the whole channel” — rung-c-specific; in the thesis world the demand-shaping channel is worth $+21.8\%$ (shed) and the compound 49% .
- “Fill 1.000” for the B960 scarcity compound — display-rounding; 0.990 (B1440 reaches 0.9999).
- The frozen per-severity buffer grids — replaced by demand-aware sizing.

5 Rigor record (one paragraph)

All experiments ran on a gated harness (bit-exact baseline reproduction on both library and CLI paths, determinism, demand conservation, cross-seed variance), with pre-registered tuning/reporting seed splits (1–10 / 11–30), dual cost accounting, and both fairness baselines carried. Every phase was audited by an independent agent re-deriving headline numbers from raw CSVs (12/12, 11/12 — the exception a display-rounding over-claim, corrected — 16/16, 13/13). Three contaminated or buggy batches were caught by gates or diagnostics (CLI-default contamination; a bootstrap deadlock; both logged) and discarded without contaminating conclusions. Simulator defects found and guarded: the normal demand model’s global RNG reseed, a dead trust-delta config constant, the silent discard of orders ≤ 1 unit, and the missing on-order timeout that produces stranded-pipeline suppression.

6 Next steps

1. **Obtain the thesis Table 3.9 configuration (incl. the base-stock baseline)** — decides the interpretation of F2’s 49%-vs-89% gap; everything else is measured.
2. Recurring-disruption robustness check before publishing the lever-flip map (premium amortization could move buffer-cell boundaries).
3. Ramped-onset regime as the second paper (the one place detection could earn value).
4. When corrected DRL code arrives: evaluate against the regime-appropriate simple compounds (thesis world: the DS-seat compound; slack: 273k; sat50: 767k–1,726k with demand-aware B), same seeds, same gates, both accounting conventions.

5. Distill the paper from this document: the three-lever value decomposition with the lever-flip map, the DS-seat result as the RL-relevant centerpiece.